EXP 16

```
# Compare Classification Algorithms

import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

# Load dataset
data = load_breast_cancer()
X = data.data
y = data.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Feature scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Classification algorithms
models = {
```

```
    "Logistic Regression": LogisticRegression(),
    "KNN": KNeighborsClassifier(),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Random Forest": RandomForestClassifier(random_state=42),
    "SVM": SVC()
}

# Evaluate models
results = []

for name, model in models.items():
    model.fit(X_train, y_train)
    pred = model.predict(X_test)

    results.append([
        name,
        accuracy_score(y_test, pred),
        precision_score(y_test, pred),
        recall_score(y_test, pred),
        f1_score(y_test, pred)
    ])

# Display results
results = pd.DataFrame(
    results,
    columns=["Algorithm", "Accuracy", "Precision", "Recall", "F1 Score"]
)

print(results.round(4))
```

OUTPUT

Algorithm	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.9737	0.9722	0.9859	0.9790
KNN	0.9474	0.9577	0.9577	0.9577
Decision Tree	0.9474	0.9577	0.9577	0.9577
Random Forest	0.9649	0.9589	0.9859	0.9722
SVM	0.9825	0.9726	1.0000	0.9861